

Appendix for Improving Paratope and Epitope Prediction by Multi-Modal Contrastive Learning and Interaction Informativeness Estimation

1 Details of MIPE

1.1 Feature construction

For structure data, we first build a 3D graph based on residue coordinates. For each node in the graph, we construct an initial 62-dimensional physicochemical feature vector as follows:

- A one-hot encoding representing residue types, with a dimension of 20.
- A position-specific scoring matrix (PSSM) obtained through PSI-BLAST computation, with a dimension of 20.
- The absolute and relative solvent accessible surface area computed by STRIDE, with a dimension of 2.
- A local amino acid profile reveals the frequency of each amino acid type within an 8Å radius of the residue, with a dimension of 20.

1.2 Construction of discriminators in Inter-CL

In multi-modal contrastive learning, we bring in an inter-CL to encourage modalities with both correct uni-modal predictions to share a stronger correspondence. To guide the model in learning effective representations for each modality, we utilize two pre-trained probabilistic discriminators for calculating their pseudo labels.

Specifically, for sequence data, we input them into a sequence encoder, obtaining sequence representations h^s . Subsequently, we feed the representation into a predictor f^s , predicting the binding probability for each residue:

$$p_i^s = f^s(h_i^s). \quad (1)$$

where f^s is the multi-layer perceptron (MLP). The supervised loss of the sequence prediction can be formulated as follows:

$$\mathcal{L}^s = -\frac{1}{M} \sum_{i=1}^M (y_i^s \log(p_i^s) + (1 - y_i^s) \log(1 - p_i^s)) \quad (2)$$

where the M is the length of the protein, y_i^s is the ground-truth label of each residue.

For structure data, similar to sequence data, we input them into a structure encoder, obtaining structure representations

h^c . We feed the representation into a structure predictor f^c to predict the binding probability for each residue:

$$p_i^c = f^c(h_i^c). \quad (3)$$

where f^c is the multi-layer perceptron (MLP). The supervised loss of the structure prediction can be formulated as follows:

$$\mathcal{L}^c = -\frac{1}{M} \sum_{i=1}^M (y_i^c \log(p_i^c) + (1 - y_i^c) \log(1 - p_i^c)) \quad (4)$$

where the M is the length of the protein and y_i^c is the ground-truth label of each residue. Finally, the overall objective function for the pre-trained discriminators is defined as:

$$\mathcal{L} = \mathcal{L}^s + \mathcal{L}^c \quad (5)$$

We employ pre-trained probabilistic discriminators, denoted as f^s and f^c , within the context of inter-CL. These discriminators are utilized to compute the pseudo labels for both sequential and structural data, providing guidance for the subsequent contrastive learning process.

2 Details of Experimental Settings

2.1 Datasets

We compile a dataset consisting of 626 binding antibody-antigen pairs, including their sequences, structures, and corresponding interaction maps. Noteworthy, antibodies primarily bind to antigens through their CDR regions. Most researchers use Euclidean distance to define paratopes and epitopes, and we follow the usual way in our dataset: within the CDR regions/antigen, a residue is labeled as a paratope/epitope if the Euclidean distance between its backbone atom and any backbone atom on the other antigen/CDR regions is less than 4.5 Å. As illustrated in Figure 1 and Figure 2, we have analyzed the length distribution of all antibodies (CDR regions)/antigens in the dataset. The average sequence length for antibodies is 77, while the average sequence length for antigens is 260.

Additionally, we have computed the percentage of binding/non-binding residues in each antibody (CDR±2 regions)/antigen, as shown in Figure 3. On average, binding residues on antibodies constitute 20.5% of all residues, while on antigens, the average percentage of binding residues is 10.6%. It is evident from these results that our dataset is characterized by class imbalance.

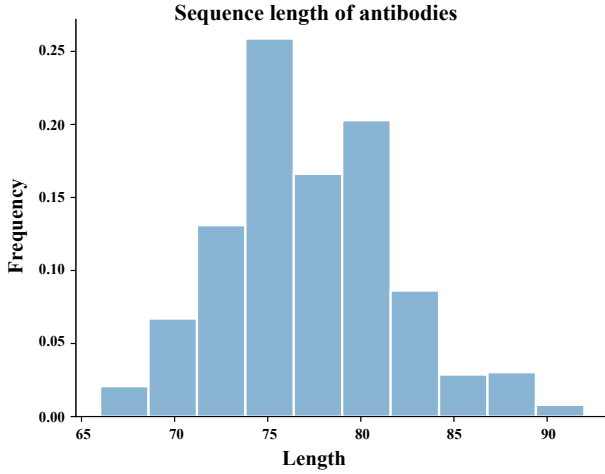


Figure 1: Distribution of antibody sequence lengths.

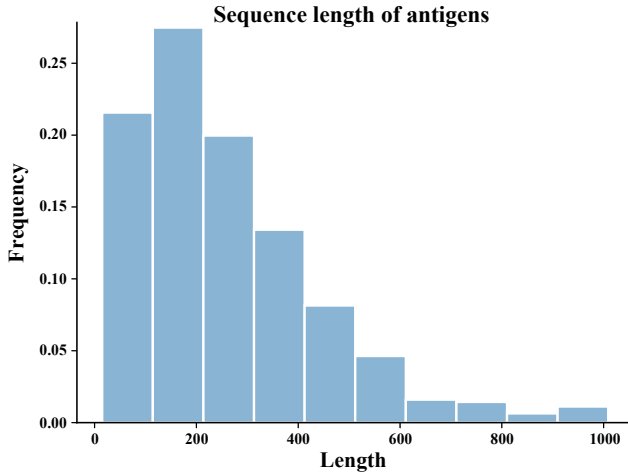


Figure 2: Distribution of antigen sequence lengths.

2.2 Baselines

For the three tasks related to paratope and epitope prediction, we compare our method MIPE with various baselines. For single paratope prediction, we choose the following methods as baselines:

- **Parapred** [Liberis *et al.*, 2018] is a model for paratope prediction, which utilizes CNNs and BiLSTMs successively to aggregate features from both the local and global antibody sequences.
- **AG-Fast-Parapred** [Deac *et al.*, 2019] is an improvement method based on Parapred, which enhances paratope prediction ability by implementing cross-modal attention over sequential antigen information.
- **Paragraph** [Chinery *et al.*, 2022] is a deep learning model, which utilizes EGNNs to extract 3D representations of antibodies, thereby predicting paratopes of the antibody.
- **PECAN** [Pittala and Bailey-Kellogg, 2020] is a context-

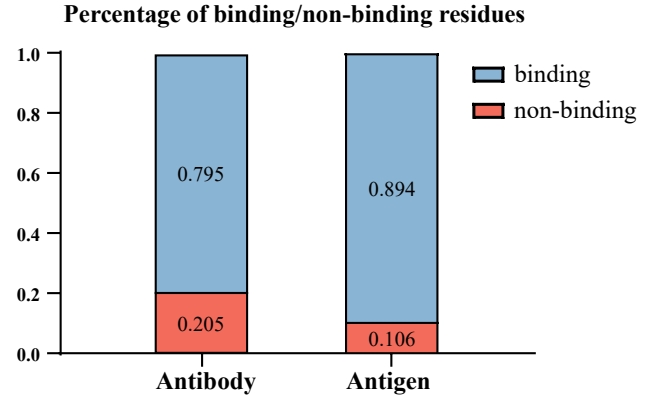


Figure 3: Percentage of binding residues on antibodies/antigens.

aware deep learning model that constructs a two-dimensional graph by utilizing given antibody and antigen structures. It encodes them separately using GCNs and leverages the interaction attention between them to predict paratopes on the antibody. Similarly, the method can also be used for epitope prediction.

- **PesTo** [Krapp *et al.*, 2023] is a parameter-free geometric deep learning method that directly encodes atomic coordinates on proteins to obtain atomic-level information and aggregates it into residue-level information. This process enables the calculation of the binding probability for each residue. Since the antibodies and antigens considered in this paper are both proteins, this method is applicable to the prediction of both paratopes and epitopes.

For sequence-based methods, Parapred, AG-Fast-Parapred, we input the antibody sequence and predict the binding probability of each residue in CDR regions. For the structure-based method, PECAN, since PECAN needs the structure of both antigens and antibodies, and according to the data collection method in PECAN, we built 62-dimensional structural features based on PDB files, input them to the model and predict the binding site of the CDR regions/antigen. For Paragraph, we use the same method as PECAN. For the protein binding site prediction method, PesTo, we input the PDB structure of antigens/antibodies and output the binding site on antigens /CDR regions.

For single epitope prediction, we choose PECAN, PesTo, and the following baselines:

- **BepiPred-3.0** [Clifford *et al.*, 2022] utilizes ESM-2 to acquire antigen representations, subsequently employing them to train various models (FFNN, CNN, BiLSTM) for epitope prediction.
- **NetBCE** [Xu and Zhao, 2022] is an interpretable deep learning model, that extracts 14 features from the antigen sequence and employs BiLSTMs and CNNs to learn the representations to predict epitopes.

For methods BepiPred-3.0, NetBCE, we input the sequence of antigens and output the binding probability of each residue.

For joint paratope-epitope prediction, we choose the following baselines:

- [Honda *et al.*, 2020] came up with a joint paratope-epitope prediction model based on a cross-attention network, it encodes the sequences of the antigen and antibody, computes the interaction attention matrix, and predicts paratopes and epitopes separately through linear layers.
- **Epi-EPMP** [Del Vecchio *et al.*, 2021] is a graph attention neural network (GAT) based method for joint paratope-epitope prediction, by encoding the structural graphs of the antibody and antigen, computing the graph attention matrix between them, and finally outputting the predicted paratopes and epitopes separately through fully connected layers.

For the sequence-based method [Honda *et al.*, 2020], we input the sequence of antigens and antibodies and predict the binding probability of each residue. The structure-based method, Epi-EPMP, adopts the same dataset construction method as PECAN, so we construct the 62-dimensional structural features according to PECAN, input them to the model and predict binding sites of the CDR regions/antigens.

2.3 Implementation

For training our model, we set the learning rate as 0.0001, the batch size as 16, and the optimizer as Adam. The experiments are performed on the machine with Intel(R)Core(TM)i9-10980XE CPU@3.00GHz and 2 GPUs (NVIDIA GeForce RTX 3090).

3 Details of performance comparison

Considering the variable lengths of antigens and antibodies, the performance of the proposed method might be influenced. Herein, we analyzed the AUC, AUPR, and MCC scores for each antibody/antigen in the independent test set (Figure 4). We observed that MIPE maintains stable performance across different lengths of antibodies/antigens.

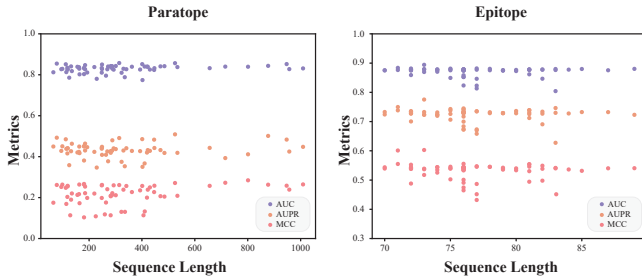


Figure 4: Evaluation scores across different sequence lengths.

4 Details of ablation study

We conducted several comparisons between MIPE and its variants MIPE (w/o CL), MIPE (w/o IIE), MIPE (w/o CL & IIE), MIPE (w/o Sequence), and MIPE (w/o Structure). We present the comparison results for single paratope prediction

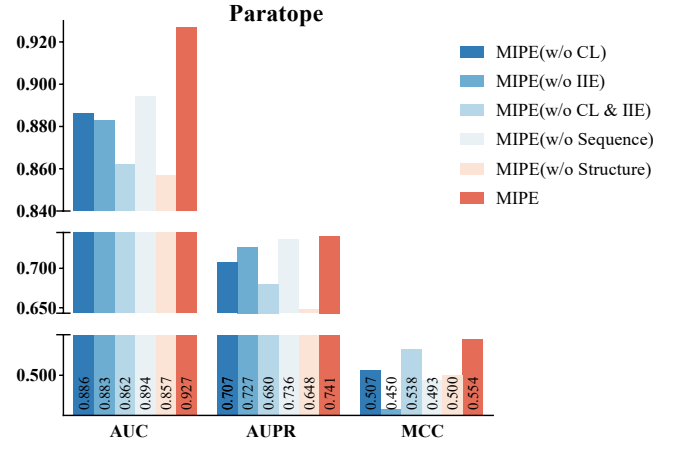


Figure 5: Results of MIPE and its variants in single paratope prediction.

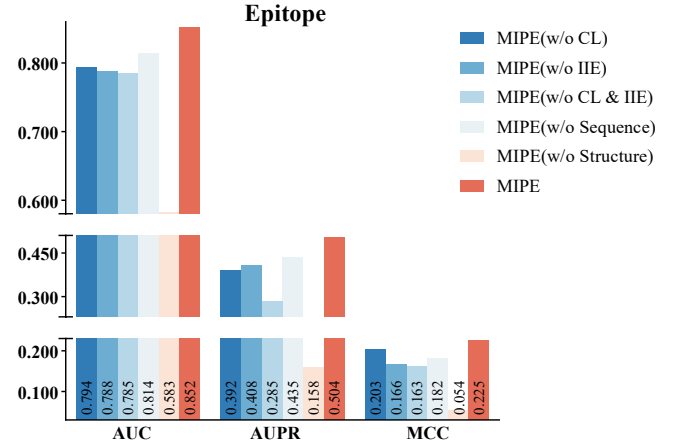


Figure 6: Results of MIPE and its variants in single epitope prediction.

and single epitope prediction in Figure 5 and Figure 6, respectively.

As shown in Figure 5 and Figure 6, the removal of the multi-modal contrastive learning model (w/o CL), the sequence encoder (w/o Sequence), and the structure encoder (w/o Structure) resulted in a noticeable decline across all metrics. These observations underscore the crucial importance of leveraging multi-modal data and aligning modalities through contrastive learning. Furthermore, the exclusion of the interaction informativeness estimation module led to a performance decrease in MIPE (w/o IIE) across all metrics. This validates the effectiveness of interaction informativeness estimation. Moreover, the performance of the variant MIPE (w/o CL & IIE) further substantiates the validity of both multi-modal contrastive learning and interaction informativeness estimation. In summary, the aforementioned results highlight the pivotal role played by these components in augmenting the model's efficacy.

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